

Subject Description Form

Subject Code	EIE2105
Subject Title	Digital and Computer Systems
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To provide students with the foundation knowledge in digital and computer systems
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Analyse and design a combinational logic circuit for practical applications; 2. Analyse and design a sequential logic circuit for practical applications; 3. Analyse the functions and features of components in a computer.
Subject Synopsis/ Indicative Syllabus	Number Systems, Operations, and Codes: Binary, octal and hexadecimal numbers; Base conversions; 1's complement and binary arithmetic; Binary-coded-decimal (BCD) representation. Combinational Logic Circuit Design: Gates, truth tables, and Boolean functions; Combinational logic design and analysis; Functional blocks and basic arithmetic logic unit. Programmable Logic Technologies: Read-only memories (ROM); Programmable logic array (PLA); Programmable array logic (PAL); Field programmable gate array (FPGA). Sequential Logic Circuit Design: Flip-flops and latches; Concept of states, Mealy and Moore machines; Sequential system analysis and design; Counters and registers. Basic Design of Computer system: Datapath and control unit; Instruction set architecture.

Teaching/ Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks		
	Lectures	1, 2, 3	Fundamental principles and key concepts of the subject are delivered to students.		
	In-Class Exercises	1, 2, 3	Some exercises and application examples are given for discussion. The students will be able to clarify concepts and to have a better understanding of the lecture material.		
	Assignments	1, 2, 3	Through working assignment and end-of-chapter problems in text books, students will develop a firm understanding and comprehension of the knowledge taught.		
	Laboratory sessions	1, 2, 3	Students will make use of the software and hardware tools to develop simple digital systems and perform simulations.		
Alignment of Assessment and Intended Subject Learning Outcomes					
	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)		
			1	2	3
	1. Continuous Assessment (total 50%)				
	• Assignments	15%	✓	✓	✓
	• Tests	25%	✓	✓	✓
	• Laboratory Exercises	10%	✓	✓	
	2. Examination	50%	✓	✓	✓
	Total	100%			
	The continuous assessment will consist of assignments, tests and laboratory exercises.				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:				
	Specific Assessment Methods/Tasks	Remark			
Assignments and tests	Assignments will be set to evaluate students’ ability in applying concepts and skills learned in the lectures. Tests will be set to challenge students to apply the course contents in a more practical setting. Students needs to think critically and creatively in order to come with an alternate solution for an existing problem.				

	Laboratory exercises	Each student is required to answer several questions related to each lab session in the lab sheet and hand in his/her answers.
	Exam	Exam will be set to evaluate students' understanding of the course contents and ability to apply them to practical questions.
Student Study Effort Required	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Laboratory/Practice Classes	12 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours
	• Laboratory/Practice Classes: preview of materials, revision and/or reports writing	33 Hours
	Total student study effort:	105 Hours
Reading List and References	Textbooks: 1. M.M. Mano and C.R. Kime, <i>Logic and Computer Design Fundamentals</i> , 5 th ed., Upper Saddle River, NJ: Prentice-Hall, 2015. Reference Books: 1. M. Rafiquzzaman, <i>Fundamentals of Digital Logic and Microcomputer Design</i> , 5 th ed., John Wiley & Sons, 2005. 2. B. Brey, <i>The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium and Pentium Pro Processor: Architecture, Programming and Interfacing</i> , 7 th ed., 2005. 3. D.A. Patterson and J.L. Hennessy, <i>Computer Organization and Design: The Hardware/Software Interface</i> , 4 th ed., Morgan Kaufmann Publishers, 2009.	